EX.NO:08 Workflow using Zoho Creator

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AIM:

To create form workflows in Zoho Creator for the SmartCare – Clinic Management System application, so that a unique patient code is generated automatically when the Patient Profiles form loads, and duplicate patient entries are cancelled when the form is submitted.

THEORY:

Workflow. A workflow in Zoho Creator is a set of actions that run automatically when a particular event occurs in an application. Workflows reduce manual effort, enforce business rules and keep the data consistent without the user having to do anything extra.

Types of workflows. Zoho Creator provides Form workflows (triggered by events on a form), Schedules (run at a set date and time), Approvals, Payments, Blueprints, Batch workflows and Functions. This experiment uses Form workflows.

Form events. A form workflow is triggered when a record is *Created*, *Edited* or *Created or Edited*, and runs at a chosen stage: *Load of the form* (actions run as the form opens), *Validations on form submission* (actions run to check field values before the record is saved), or *On success* (actions run after the record is saved).

Deluge scripting. Actions in a workflow are written in Deluge (Data Enriched Language for the Universal Grid Environment), the scripting language of Zoho Creator. It supports conditions, variables, built-in functions such as `randomNumber()`, `toString()` and `concat()`, system variables such as `zoho.currentdate` and `zoho.loginuserid`, and record fetching using a criteria such as `Form_Name[field == value]`.

Form load workflow. The workflow *Auto-generate Patient Code* runs on the load of the form. It checks whether the record is new, builds the code as *prefix + date (yyyyMMdd) + six-digit random number* and assigns it to the `patient_code` field, so each patient gets an identifier without manual entry.

Validation workflow. The workflow *Cancel Duplicate Patient Entry* runs on validations on form submission. It fetches records whose patient code matches the entered one; if the count is greater than zero, the `alert` statement shows an error message and `cancel submit` stops the record from being saved, thus preventing duplicate patient records.

PROCEDURE:

1 Sign in to Zoho Creator and open the SmartCare – Clinic Management System application.

2 Click the **Workflow** tab at the top and select **Form workflows**. The forms of the application (Patient Profiles, Appointment Scheduling, Apply Leave, Billing Invoices, Medical Records and Staff Directory) are listed on the left.

3 Select the **Patient Profiles** form and click **New Workflow**.

4 Name the workflow **Auto-generate Patient Code**. Set the event to **Created** and choose **Load of the form** as the stage, then click Create.

5 Under *Execute without condition*, click **Add New Action**, select **Deluge script** and open the script editor.

6 Write the script: if `input.ID` is null or 0 (a new registration), generate a six-digit random number using `randomNumber(100000,999999);` get the current date as a *yyyyMMdd* string using `zoho.currentdate` and `toString();` get the logged-in user's email using `zoho.loginuserid` and take its first three characters as the prefix (default prefix "PAT" otherwise).

7 Combine the parts using `concat()` to form the patient code and assign it to `input.patient_code`. Save the script and click **Done**.

8 Create a second workflow on the Patient Profiles form named **Cancel Duplicate Patient Entry**. Set the event to **Created or Edited** and the stage to **Validations on form submission**.

9 Add a **Deluge script** action. In the script, check whether `input.patient_code.isNull();` if it is empty, allow normal submission.

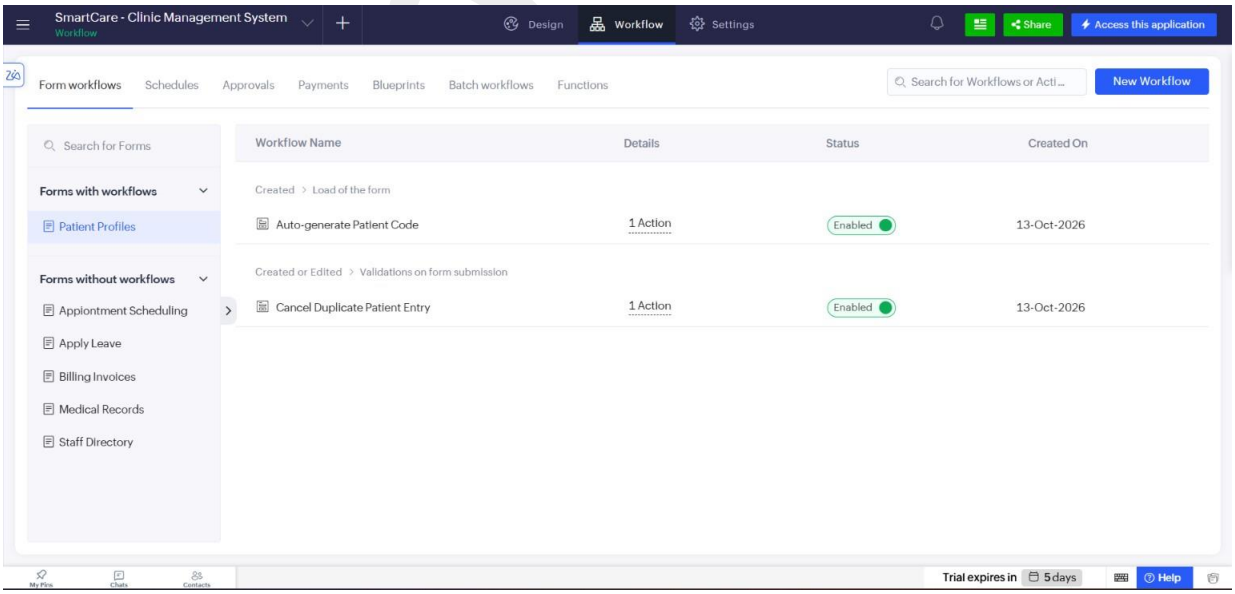
10 Otherwise, fetch the matching records using `Patient_Profiles[patient_code == input.patient_code]` and check whether `count()` is greater than 0. If so, display the alert "Error: Patient code … already exists" and use `cancel submit;` to stop the submission.

11 Save the script, click **Done**, and make sure both workflows show the status **Enabled** in the workflow list.

12 Click **Access this application**, open the Patient Profiles form, and verify that the patient code is auto-generated on load and that a duplicate patient code is rejected on submission.

OUTPUT:

Fig. 1: Form workflows of the SmartCare application (both workflows enabled)



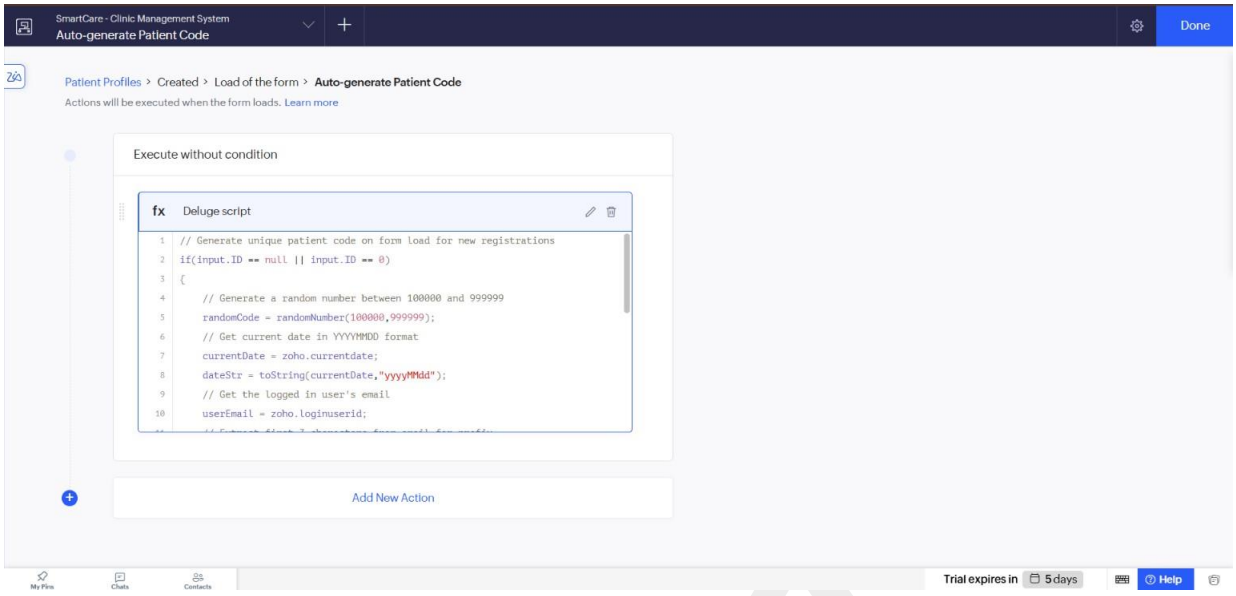


Fig. 2: Auto-generate Patient Code – Deluge script (start of script)

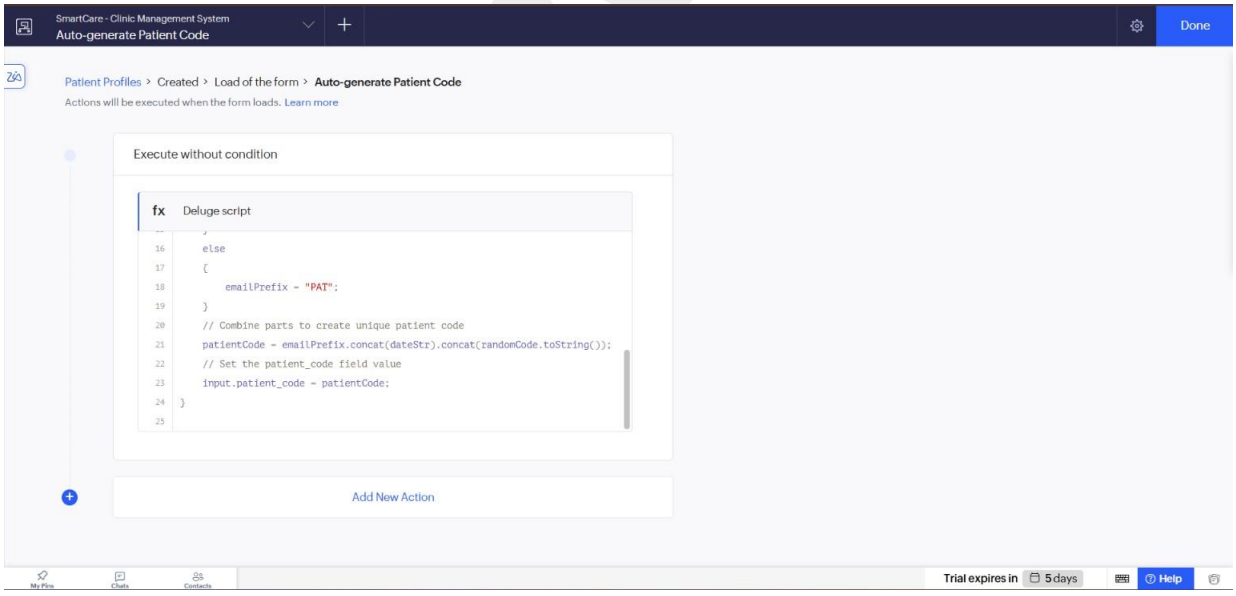


Fig. 3: Auto-generate Patient Code – Deluge script (patient code creation and assignment)

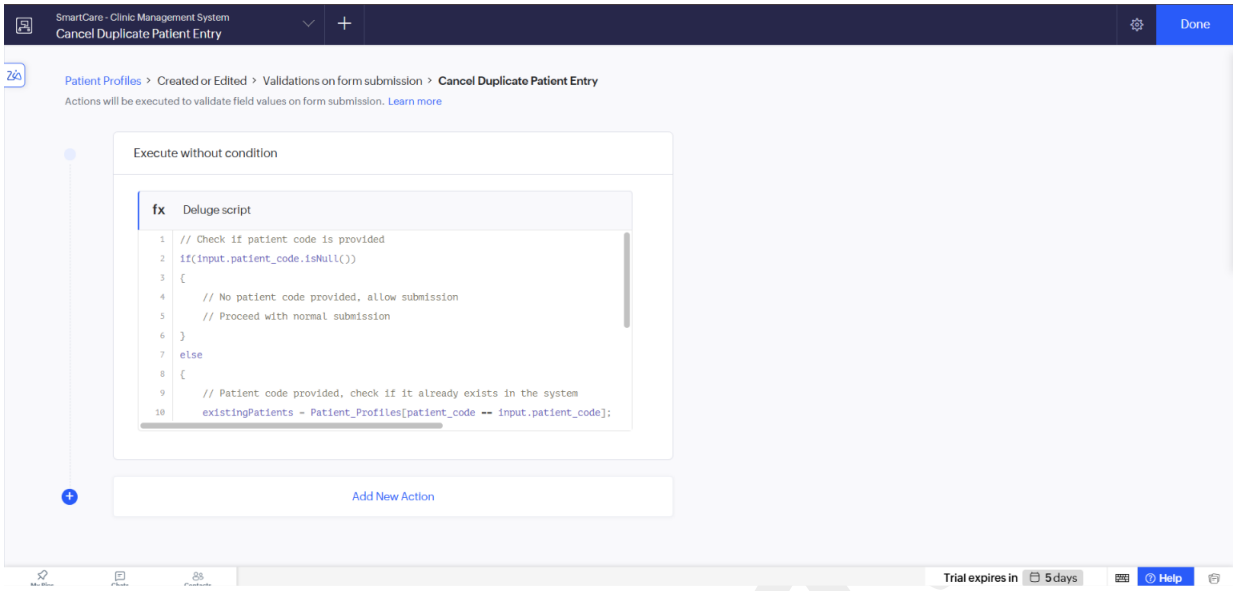


Fig. 4: Cancel Duplicate Patient Entry – Deluge script (check for patient code)

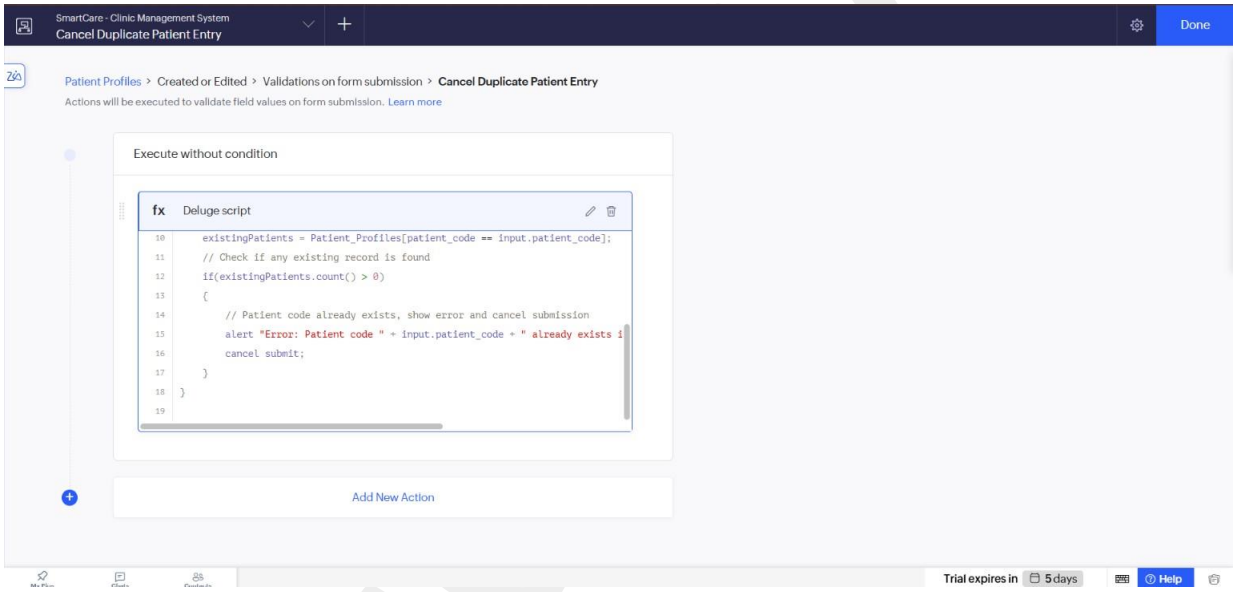


Fig. 5: Cancel Duplicate Patient Entry – Deluge script (duplicate check, alert and cancel submit)

RESULT:

Thus, the form workflows *Auto-generate Patient Code* and *Cancel Duplicate Patient Entry* were successfully created using Deluge scripts in Zoho Creator for the SmartCare – Clinic Management System application. A unique patient code is generated automatically when the form loads, and duplicate patient entries are cancelled on submission.